SECTION 11 MISCELLANEOUS ROADWORKS

PART 3: PEDESTRIAN GUARD-RAILING

MATERIALS

Stainless steel 11.29***S*** Stainless steel for pedestrian guard-railing shall comply with the following:

- Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes : BS EN 10088-2
- Stainless steel tubes suitable for threading in accordance with BS 21 : BS 6362
- Wrought stainless steel : BS EN 10084, BS EN 10085, BS EN 10087, BS EN 10095, BS EN 10250-4 and PD 970, Grade 1.4401.
- Flat rolled stainless steel : BS EN 10029, BS EN 10048, BS EN 10051 + A1, BS EN 10095 and BS EN ISO 9445, Grade 1.4401 in the softened condition.
- Stainless steel tubes : BS EN 10296-1 and BS EN 10296-2, designation LW 23 GZF(S).
- Stainless steel bolts and nuts : BS EN ISO 3506- 1&2, steel Grade A4 and property class 80.
- Stainless steel washers : BS EN 10029, BS EN 10048, BS EN 10051 + A1, BS EN 10095 and BS EN ISO 9445, Grade 1.4401 in the softened condition.

Bolts, nuts, screws, washers and rivets 11.31 (4)***S*** Galvanized bolts, nuts, screws, washers and rivets shall be used with galvanized pedestrian guard-railing unless shown otherwise in the Contract. Non-ferrous or stainless steel components shall be insulated from ferrous materials by non-conductive insulators of a type approved by the *Supervisor*.

Not Used 11.32A (1) Not Used

FABRICATION OF PEDESTRIAN GUARD-RAILING

Galvanizing to steel 11.33 (1)***S*** Steel components forming pedestrian guard-railing shall be hot-dip galvanized in accordance with BS EN ISO 1461: 1999 to a coating thickness of at least:

(a) 500g/m^3 for railing other than on highways structures;

(b) 600g/m^3 for highways structures.

Hole opening 11.34A All necessary hole openings shall be drilled before galvanizing.

PART 4: UNTENSIONED BEAM BARRIERS

MATERIALS

Beams 11.39 (7) Untensioned corrugated beam barrier fences shall be to the details given on the Highways Department Standard Drawings H2112B, H2113A, H2114B, H2115A, H2116A, H2117D, H2119E, H2120A, H2121, H2122, H2123A, H2124B, H2125B, H2126A, H2127B, H2128A and H2129C inclusive or any latest update.

(8) Beams shall be shaped so as to present no sharp edges to traffic and except where shown on the Drawings shall be straight and of uniform cross-section within manufacturing tolerances.

Blocking-out 11.39A (1) The beams shall be separated from the posts by blocking-out pieces which are fabricated from materials of the same quality as the posts.

(2) Posts and blocking-out pieces shall be drilled to take the specified post bolts for affixing the beams.

Posts 11.40 (2)***S*** Posts and blocking-out pieces shall be hot-dip galvanized in accordance with BS EN ISO 1461: 1999 to a coating thickness of at least 610 g/m^2.

(5) Channel sections, or other sections with exposed sharp corners which may be potentially harmful to pedestrians, shall not be used as posts for untensioned beam barriers installed on footways or cycletracks. For posts fabricated from hollow sections, the top sealing plate shall be rounded at all edges and corners.

CONSTRUCTION OF UNTENSIONED BEAM BARRIERS

Installation of untensioned beam barriers 11.47 (2)***S*** The horizontal alignment of fences shall not depart from the road alignment by more than (+ or -) 30 mm nor deviate from the straight by more than 12 mm in any two section lengths. The horizontal centre line of the beam shall be 600 mm above the edge of the adjacent carriageway or above the datum level shown on the Drawings, within a tolerance of (+ or -) 30 mm. In addition the deviation from the straight shall not exceed (+ or-) 6 mm in any two section lengths.

(3A) Adjacent beams shall be connected by lap joints made in the direction of the adjacent traffic movement to avoid vehicles striking beam ends. Bolts are to be so located that they present no projection to traffic.

(5) Unless otherwise stated, beams shall be fixed to posts by blockout mountings.

(6) Posts with concrete footings shall be used for the first post nearest an anchorage in poor ground, or if the clearance between the traffic face of the fence and the face of an obstruction is less than 1m, or if the fence is installed on footways, unless otherwise stated in the Contract.

Concrete footings 11.49 (4) Concrete footings to posts shall be in accordance with HyD Standard Drawing H2119E. They shall not protrude more than 75 mm above the surrounding ground. Where, in locally low ground, more than 75 mm of concrete would have been exposed, in order to maintain the line of the fence the concrete footing shall be lowered and a longer post installed such that the length of post in the footing is not reduced to less than 320 mm. The exposed concrete shall be sloped from the post to ground level by about 1 in 10. Where these conditions cannot be met, the ground level may be raised locally to suit, subject to the *Supervisor*'s Approval.

(5) Concrete footings shall be used for untensioned beam barriers installed on footways or cycletracks.

Stability of post foundations 11.49A (1) The foundation for the posts shall be able to resist an overturning moment of 6000 Nm with a deflection of 100 mm. Where there is doubt as to the adequacy of the post foundations, the *Supervisor* may instruct site testings of the ground where posts are to be installed.

(2) The following test shall be carried out to determine suitability of post foundation:-

(a) The test post shall be a 102 mm x 102mm x 14.53Kg/m mild steel tee-section. The distribution plate is welded onto the tee flange of the driver post and the post is driven vertically into the ground. Where posts are set in concrete or where it is not possible to drive the tee-section test post, the test may be carried out on installed posts by the use of a sleeve placed over the posts. For concrete footings, the post shall be either mounted in a socket or bolted to the top of the footing using the standard bridge deck post fixings.

(b) When a horizontal load is applied at a convenient height and at right angles to the tee flange, a bending moment of 6000 Nm shall be achieved at ground level before deflection of the post exceeds 100 mm measured at a distance of 600 mm above ground level. Where this criterion cannot be achieved, a long driver post will be

required, unless it is considered that the deflection requirement would still not be met because of poor ground conditions, in which case, concrete foundations should be used.

Anchor blocks 11.50 (3) All untensioned corrugated beam barrier fences, unless connected to bridge parapets, shall be ended to a concrete anchor block.

(4) Concrete anchor blocks shall be cast directly against the side of the excavation except where there is excessive overbreak, in which case the overbreak shall be made good with grade 10 concrete.

PART 5 : KERBS, EDGINGS AND QUADRANTS

MATERIALS

Concrete kerbs, edgings and quadrants 11.51 (3) Precast concrete kerbs, edgings and quadrants to HyD Standard Drawings H1118 & H1119 used in the *works* shall generally be those manufactured by the Correctional Services Department (CSD). The *Contractor* shall inform the *Supervisor* of the quantities required 4 months in advance before the collection of kerbs and the *Supervisor* may place the order with the CSD. On notification of their availability the *Contractor* shall make arrangements to collect them form the Tai Lam Correctional Institution or other Correctional Institutions as directed by the *Supervisor* and store them at the Site.

(4) Since the precast concrete kerbs, edgings and quadrants are supplied by the CSD, the *Contractor* shall be charged the prices obtainable from CSD. Such charge will be recovered by appropriate deductions from money due to or becoming due to the *Contractor* and shall be excluded from the "Effective Value" for the calculation of Contract Price Fluctuations.

(5) The *Contractor* shall as soon as practicable after Contract Date submit to the *Supervisor* his programmed requirements for acquisition of kerbs. When requested by the *Contractor* and having regard to the progress of the *works* the *Supervisor* shall issue requisition orders from time to time. The *Contractor* will thereafter be responsible for agreeing with the superintendent of Tai Lam Correctional Institute a suitable date and time for collection of the kerbs, edgings and quadrants.

(6) If excess kerbs, edgings and quadrants are required from Government as a result of wastage, wrong method of working, defective works and damages etc., the *Contractor* shall reimburse all costs to Government. The *Contractor* shall also be responsible for all additional expenses incurred in labour, transport etc. for obtaining these kerbs.

(7) Should, however, the CSD be unable to supply the order by the date requested for collection, the *Supervisor* will instruct

the *Contractor* two months prior to this date, to supply the ordered kerbs from his own source. Adjustment items in the Activity Schedule shall be included for this eventually.

CONSTRUCTION OF KERBS, EDGINGS AND QUADRANTS

Construction of precast concrete and granite kerbs, edgings and quadrants

11.53 (4) Expansion joints in kerbs shall be formed at not more than 20m centres and shall consist of a 26mm wide approved joint filler continuing through the kerb foundation and backing with a 25mm wide by 20mm deep approved joint sealant.

(5) For laying of precast concrete kerbs along a straight alignment, the kerbs are normally laid adjoining each other, and therefore joints between kerbs should not be filled and flushed with cement mortar.

(6) For laying of precast concrete kerbs along a sag curve, a crest curve or a corner where radius kerbs and quadrants as shown in HyD Standard Drawing H1118 are not applicable, joints between kerbs shall then follow the requirements as stipulated by Clause 11.53(2).

(7) For laying of granite kerbs, the requirements as set out in Clause 11.53(2) should always be followed because of the inherent irregularities on stone surface.

PART 6 : FOOTWAYS, CYCLETRACKS AND PAVED AREAS

CONSTRUCTION OF FOOTWAYS, CYCLETRACKS AND PAVED AREAS

In-situ concrete footways, cycletracks and paved areas

11.57***S*** (1) In-situ concrete for footways, cycle tracks and paved areas shall be laid in areas not exceeding 20 m2. The finish to the concrete surface shall be Class U4.

(2) Footway joints shall be open joints of 3mm to 6mm wide and 20mm deep and shall be sealed in accordance with Clause 10.40.

(3) Expansion joints shall be formed at 20 metre centres. The grooves shall be 20mm wide by 55mm deep and shall be sealed in accordance with Clause 10.12.

(4) The *Contractor* shall strictly follow the guidelines "HQ/GN/05A - Guidelines on Good Practice for Laying Unit Pavers" as promulgated by the Highways Department.

Resin based coloured dressing

11.58A (1) The coloured dressing shall consist of a resin-based binder sprayed on to a sound substrate and covered with coloured aggregates to form a coloured dressing.

(2) The coloured aggregate can be naturally occurring colour aggregates or pigmented aggregates formed by the coating of an ultra-violet resistant pigment to the aggregates using epoxy or other approved binder.

(3) The road surface shall be prepared to the satisfaction of the *Supervisor* before application of coloured dressing which shall also be cured sufficiently before opening to traffic.

(4) The aggregates shall be calcined bauxite with PSV ≥ 65 and AAV ≤ 5 or other aggregates to be approved by the *Supervisor*. The grading of the aggregates shall be between 0.6mm and 1.4mm to provide a desired texture.

(5) The dressing has to achieve a minimum skid resistance of 45, when applied to carriageways which are level and straight.

(6) On a smooth close texture carriageway surface, the amount of binder shall not be less than 1.35kg/m^2 or such rate as specified by the resin manufacturer. On more rugged surfaces, a greater rate of spread is required.

(7) For a period of 4 years from the date on which a certificate of completion of the *works* shall have been issued, there shall be no loss of adhesion of the finished coating to the road surface. Any defects shall be made good at the *Contractor*'s own expense.

PROTECTION OF FOOTWAYS, CYCLETRACKS AND PAVED AREAS

Cobble-stone paving 11.59A Cobble-stone paving shall consist of stones placed in a 100 mm thick grade 30/20 concrete bed. The stones shall be roughly elliptical and have a maximum dimension not exceeding 150 mm and a minimum dimension not less than 100 mm. The stones shall be placed in the concrete bed to such a depth that the nominal line of the horizontal diameter of the stone is about 10 mm below the finished concrete level. The clearance between each stone shall be 75 mm maximum and 40 mm minimum. Cement wash shall not coat the exposed surface of the stones. Before the *Contractor* places any cobble-stone paving in the permanent works he shall prepare, for the agreement of the *Supervisor*, a 2 m square trial panel of finished cobble-stone paving.

Rubble stone facing and paving 11.59B

(1) Rubble stone facing shall consist of random stone whose maximum dimension on the exposed face shall not be less than 300 mm, whose minimum dimension on the exposed face shall not be less than 200 mm and whose thickness shall not be less than 200 mm. Stones shall be carefully shaped to obtain as close a fit as possible at beds and joints. When placed on the ground the stones shall be placed on material trimmed and compacted to make a firm bed.

(2) Rubble stone paving shall consist of stones of similar size to those used in rubble stone facing. Stones shall be bedded on 100mm of concrete grade 20/40 and sufficient space shall be left between stones to allow the joints to be filled completely with concrete or mortar. The stones shall be bedded on the concrete while it is still fresh, or alternatively the concrete shall be left with a rough finish and the stones shall be bedded on a thin layer of mortar after the concrete has set. No cavities shall

be left under or between the stones.

PART 7 : PRECAST CONCRETE UNITS FOR PAVING

MATERIALS

Concrete and use of recycled aggregates

11.65***S*** (1) Concrete for units in footways and cycle tracks shall be Grade 30. Concrete for units in carriageways or areas to which vehicles will have access shall be Grade 45.

(2) The nominal maximum aggregate size for concrete in precast units shall be 10 mm.

(3) Aggregates for concrete shall comply with the following requirements:

(a) The aggregates shall contain not less than 70% by weight of recycled aggregates.

(b) The recycled fine aggregates shall constitute not less than 40% by weight of the total recycled aggregates.

(c) Coarse recycled aggregates shall be retained on a 5mm BS test sieve.

(d) Fine recycled aggregates shall all pass a 5mm BS test sieve.

(e) The recycled aggregates shall contain not more than 0.5% of wood and other materials less dense than water by using the manual sorting test method in accordance with BRE Digest 433.

(f) The recycled aggregates shall contain not more than 1% of other foreign materials (e.g. metals, plastics, clay lumps, asphalt and tar, glass, etc.) by using the manual sorting test method in accordance with BRE Digest 433.

(g) The recycled aggregates shall be the crushed inert construction and demolition materials produced from the crushing plant of the Civil Engineering and Development Department or other local sources approved by the *Supervisor*.

(h) The nominal maximum aggregate size of aggregates shall be 10 mm.

(4) Notwithstanding sub-clause (3) above, the *Contractor* may propose for the *Supervisor*'s approval the use of virgin aggregates in lieu of recycled aggregates in the concrete when there is a shortage of supply of recycled aggregates.

SUBMISSIONS

Particulars of paving units

11.68 (1) (e) Where recycled aggregates are not used, the *Contractor* shall submit the following additional particulars to the

Supervisor:

- a written confirmation from the manufacturer confirming the reasons why recycled aggregates are not used, e.g. shortage in supply of recycled aggregates in the market; and

- proposed quantities of paving units for which recycled aggregates are not used.

PART 8 : MISCELLANEOUS

CONSTRUCTION

Use of surfaces by vehicular traffic

11.89 (1) Vehicular traffic on pavements under construction shall be limited in relation to the capacity of the pavement courses so that damage is not caused to the formation or the material laid.

(2) The various pavement courses shall be kept free from deleterious materials at all times.

(3) Cleaning of contaminated surfaces shall be carried out as instructed by the *Supervisor*.

(4) Vehicular traffic shall not be permitted, on surfaces as follows:

(a) On concrete slabs until a minimum period of at least seven days has elapsed after completion and until all joints and grooves have been sealed, or as approved by the *Supervisor*.

(b) On bituminous material courses, until at least 6 hours after laying. This may be reduced with the approval of the *Supervisor* but only if the material has cooled sufficiently to resist deformation under the traffic.

Services access covers, frames, and street hardware

11.90 (1) Flexible Road Construction

(a) Before any work commences on the construction of the carriageway, the installation of drains, sewers and other underground services shall be completed including backfilling and compacting up to the bottom of the roadbase.

(b) All service access covers, frames and similar hardware that would prevent continuous paving operations shall be omitted until such operations are completed, unless directed by the *Supervisor*.

(c) The *Contractor* shall use temporary covers and displacers, such as steel plates, to achieve the covering of the access chambers and these shall be strong enough to withstand the construction traffic loading.

(d) On completion of laying operations, the materials forming the roadbase and basecourse shall be cut out, the

temporary covers removed and the covers, frames and hardware set in place to the correct line and level. Backfilling around the covers, frames or hardware shall be completed with a suitable bituminous material complying with Table 9.3 of GS. Compaction up to the underside of the wearing course or friction course, shall be performed using handrammers or other suitable mechanical equipment, in layers not exceeding 50 mm.

(e) Wearing course or friction course operations shall proceed after covers, frames and all other hardware have been set to the correct level. The finished surface level of the covers, frames and other hardware shall be neither below nor more than 5 mm above the level of adjoining pavements, except that gully gratings shall be neither above nor more than 5 mm below the level of adjoining pavements.

(f) Notwithstanding sub-clauses (d) and (e), the temporary covers shall only be removed and the covers, frames and hardware set in place to the correct line and level after completion of the uppermost layer of pavement when instructed by the *Supervisor*.

(2) Concrete Road Construction

In the case of concrete road construction, the installation of service access covers, frames and other street hardware shall generally be in accordance with the methods outlined above, except that installations shall be completed, inclusive of backfilling and compaction before placement of the pavement slab.

CRASH CUSHION SYSTEM

Crash cushion system 11.91 (1) The crash cushion system shall be of a proprietary type approved by the *Supervisor* and shall comply with the requirements of this Clause. All steel components shall be hot-dip galvanized. Prior to approval of the crash cushion system, the *Contractor* shall submit the draft form of warranty for the approval of the *Supervisor*. In addition, the *Contractor* and the manufacturer shall jointly submit an undertaking for the manufacturer to provide the warranty in the approval form to the *Supervisor* and HyD/NT in consideration of the crash cushion system being approved for use.

(2) General performance characteristics

The crash cushion shall be able to withstand head-on, side-angle and reverse-angle impacts up to the design speed specified for the location. The design speed refers to the specified maximum travelling speed the crash cushion can withstand. Additionally, the crash cushion shall be able to satisfy all of the following performance characteristics:

(a) Non-gating

The crash cushion shall not allow the impacting vehicle to "gate" or pass through the crash cushion when it is struck at an angle on the front or "nose" to avoid the potential for secondary impacts.

(b) Redirective

The crash cushion shall redirect vehicles in all design side impacts on the unit at angles not exceeding 20° back to the originally travelled direction. Further, the redirected vehicles shall leave the crash cushion at no greater than 60% of the impact angle to avoid the potential for secondary accidents with vehicles travelling in adjacent travel lane.

(c) Non-snagging

The design of the crash cushion shall be free from any protruding elements which may cause an errant vehicle to "snag" during redirection due to design impacts into either side of the crash cushion. Such snagging is undesirable as it may cause the vehicle to change direction in an uncontrolled phenomenon which will increase the potential for secondary accidents with other travelling vehicles.

(d) Anti-climb characteristics

The crash cushion shall possess anti-climb characteristics to ensure that impacting vehicles will not vault over the system during an impact.

(e) Climactic compatibility

The system shall be suitable for use in the typical climatic conditions in Hong Kong. The energy absorbing units and other components of the crash cushion shall not be subject to damage or performance deterioration due to prolonged exposure to high ambient temperatures, high ambient humidity or frequent heavy rains.

(f) Design adaptability

The length and width of the system shall be adaptable so that it may be designed to fit within the area allocated for the crash cushion at the prescribed site.

(g) Debris

Debris from the units shall be retained within the crash cushion after either head-on, side-angle or reverse-angle design impacts. The system shall be designed such that no parts fly loose on impact.

(h) Colour of nosing and retroreflective sheeting on nosing

The colour of the nosing of the crash cushion system shall be plain yellow. Yellow retroreflective sheeting complying with ASTM D4956-99 Type IV material shall be stuck on the nosing unless otherwise approved by the *Supervisor*.

(3) Testing criteria

The crash cushion system shall comply with the testing requirements given in the United States National Cooperative Highway Research Programme (NCHRP) Report 350 in relation to structural adequacy, occupant risk and post impact vehicular trajectory except that passing criterion F is also required for level 1 test for occupant risk in addition to other requirements in the NCHRP Report 350. The contractor shall submit documentary evidence from the manufacturer, including certified test reports if required, to demonstrate compliance with NCHRP Report 350 to the satisfaction of the *Supervisor*.

(4) Storage of crash cushion

All elements and components of the crash cushion system shall be stored properly under a cover that will effectively protect them from rain. While in storage, there should be a minimum 150mm of space between the lower most element and the ground surface to avoid direct contact with soil. The *Contractor* shall comply with any direction from the manufacturer regarding proper storage of the system.

(5) Minimum stock of spare parts

The *Contractor* shall keep a minimum stock of spare parts of the crash cushion system as ordered by the *Supervisor* for future replacement of parts after crash impact.

(6) Installation

The *Contractor* shall submit for the approval of the *Supervisor* a complete set of drawings, procedures, instructions and the like which detail the steps of the installation of the proposed crash cushion. All installation works shall be conducted in accordance with the submitted procedures approved by the *Supervisor*.

(7) Setting out

The alignment and finished levels of the crash cushions shall be set accurately on site before commencement of installation work.

(8) Supervision

The crash cushions installation works shall be conducted under the direct supervision of a qualified professional supplied or approved by the system manufacturer. The *Contractor* shall submit to the *Supervisor* the qualifications of the nominated supervisor who has been fully trained to discharge the assigned task or has obtained at least 3 years' experience in crash cushion installation works in a supervisory position. If the *Supervisor* is not satisfied with the proposed supervisor, he has the right to reject the nominated supervisor and request the *Contractor* to propose another person to supervise the system installation. The *Supervisor* can instruct the *Contractor* to suspend the installation work in the event that work is undertaken in the absence of the approved supervisor. Any cost incurred as a result of such suspension of works shall be borne by the *Contractor*.

(9) Foundations and anchorages

The foundations and anchorages required to ensure the permanent positioning of the crash cushions shall be constructed in accordance with the manufacturer's drawings and procedures approved by the *Supervisor*. In addition, all excavation, erection of formwork, placing of reinforcement and concrete for foundations shall be carried out in compliance with the Specifications.

MOVABLE BARRIER FOR EMERGENCY OPENING

Movable barrier 11.92 (1) Movable steel barrier (MSB) shall comply with the evaluation criteria for Tests 3-10, 3-11 and 3-21 stipulated in Table 3.1 of the National Cooperative Highway Research Program Report 350 (NCHRP Retort 350). The maximum permanent deflection of the barrier system in the following tests shall not exceed a value that renders the barrier to encroach upon the opposite traffic lane after impact.

(a) Tests 3-11 & 3-21 of NCHRP Report 350 for roads of posted speed limits above 70 km/h

(b) Tests 2-11 & 2-21 of NCHRP Report 350 for roads of the posted speed limit of 70km/h

(c) Tests 1-11 & 1-21 of NCHRP Report 350 for roads of posted speed limits not exceeding 50 km/h

(2) The *Contractor* shall provide details of the proposed MSB and a copy of the relevant crash test certificates for approval of the *Supervisor* before delivery of the barrier system or commencement of site work, whichever is the earlier.

(3) If no MSB in compliance with the above deflection requirement could be identified from the market for an opening at an extraordinarily narrow central reserve, consideration should be given to relocation of the opening. If relocation of the opening is not feasibIe, a MSB with the least permanent deflection when tested according to sub-clause (1).

above may be considered. Provided that the predicted amount of encroachment upon the opposite traffic lane under impact conditions stipulated in sub-clause (1). above does not exceed 350mm, the *Supervisor* may approve the use of the product.

(4) The MSB shall be able to be opened by swinging about either ends towards either sides of a road smoothly up to 45 degrees on site (taking into account the longitudinal and transverse gradients of the carriageway and the central reserve), and then closed properly without affecting its containment capability. After the MSB is swung opened, the passageway created along the central reserve should be 8m to 10m long.

(5) Opening together with closing of the MSB shall be able to be completed within 2 minutes by 2 regular Asian adults on site by hands with a non-powered mechanical system (use of compressed air or any other powered means is not allowed) built-in with the MSB (e.g. a hand operated jacking system) without the need of any other tool. The system shall also have a built-in mechanism to prevent it from sliding when being opened or closed on site.

(6) If the proposed MSB is not a type which has been agreed by TD, FSD and HKPF for installation at emergency crossing (EC) previously, the *Contractor* shall solicit agreement of these traffic and rescue authorities for the use of the barrier system in advance.

(7) The *Contractor* shall arrange training session for the regional staff of FSD and HKPF to practise operation of the MSB on site before commissioning of each new MSB. Training session shall be arranged no matter the proposed MSB has been agreed for use at other ECs before or not, unless otherwise agreed by FSD and HKPF.

(8) The *Contractor* shall provide detailed design of the reinforced concrete end anchors for the MSB in consultation with its manufacturer taking into account the actual site conditions. The *Contractor* shall demonstrate that the design of the anchor is agreed by the manufacturer of the MSB, and submit the design to the *Supervisor* for approval.

(9) The *Contractor* shall inspect the proposed site before carrying out detailed design for the end anchor blocks. The *Contractor* shall identify necessary modification to existing features and installations on site and submit modification proposal to the *Supervisor* for approval.

(10) The protruding ends of any bolts on a MSB or its anchors which are exposed to traffic shall be capped by dome shape headed nuts.

(11) The *Contractor* shall submit the installation manual and the maintenance manual of the MSB at least 7 calendar days before commencement of the installation of the MSB.

(12) The *Contractor* shall provide detailed mechanical drawings of the MSB and as-built drawings for the *works* together with a detailed schedule of accessories and components for the MSB.

CODE OF PRACTICE FOR THE LIGHTING, SIGNING AND GUARDING OF ROAD WORKS ON EXPRESSWAYS

Code of Practice for the Lighting, Signing and Guarding of Road Works on Expressways

11.93 The *Contractor* shall follow the Code of Practice for the lighting, signing and guarding of road works on expressway, which prevails over those in the Code of Practice for the Lighting, Signing and Guarding of Road Works (5th Issue, 2017).